Roll No. ....

TEE–302

B. TECH. (EE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ELECTRICAL MACHINES—I

Time : 1½ Hours

Maximum Marks : 50

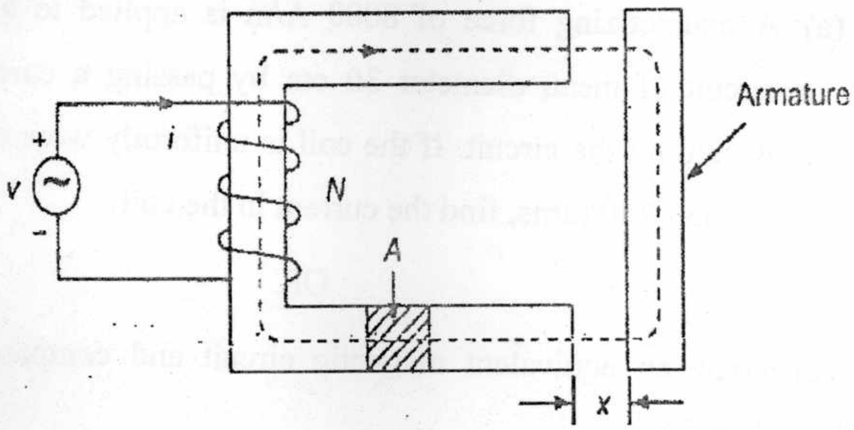
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) In the electromagnetic relay shown in figure below excited from a voltage source, the current i and flux linkages λ are related as :

$$i = \lambda^2 + 2\lambda(1-x)^2; \quad x < 1$$

Find the force on the armature as a functions of λ. (CO4)



OR

(b) Describe multiply excited magnetic field system and hence find torque in a doubly exited system. (CO4)

2. (a) Define co-energy and energy with suitable diagram. (CO2)

OR

(b) Explain the equation of force that governs singly excited magnetic field. (CO2)

3. (a) A mild steel ring has a radius of 50 mm and a cross-sectional area of 400 mm^2. A current of 0.5 A flows in a coil wound uniformly around the ring and the flux produced is 0.1 mWb. If the relative permeability at this value of current is 200, find (i) the reluctance of the mild steel and (ii) the number of turns on the coil. (CO1)

OR

(b) Discuss self and mutual inductance with suitable diagram considering doubly excited system. (CO1)

4. (a) A magnetizing force of 8000 A/m is applied to a circular magnetic circuit of mean diameter 30 cm by passing a current through a coil wound on the circuit. If the coil is uniformly wound around the circuit and has 750 turns, find the current in the coil. (CO3)

OR

(b) Draw an equivalent magnetic circuit and compare it with electrical circuit. (CO3)

5. (a) The maximum working flux density of a lifting electromagnet is 1.8 T and the effective area of a pole face is circular in cross-section. If the total magnetic flux produced is 353 mWb, determine the radius of the pole face. (CO3)

OR

(b) Discuss B-H characteristic. Why is the operating point not selected in the saturation zone of the B-H characteristic ? (CO3)